

M002 Week 3 — English Worksheet

Topic: AND & OR — Smarter Decisions · **Course:** Maze Madness · **Time:** about 45 minutes

This week your agent checks **two things at once** using **AND** and **OR**.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
if wall ahead AND wall on right:
    turn left
```

Both conditions must be true. When does the agent turn left? When does it do nothing?


```
if no wall ahead OR no wall below:
    move forward
```

Only one condition needs to be true. When does the agent move?


```
keep doing forever:
    if wall ahead AND wall on right:
        turn left
    otherwise if wall ahead:
        turn right
    otherwise:
        move forward
```

The agent reaches a corner with walls on the front AND right. What does it do?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to turn left **only when** there is a wall ahead AND a wall on the right (a dead-end corner).

```
if wall ahead OR wall on right:  
    turn left
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to move forward when **either** the path ahead is clear OR the path below is clear.

```
if no wall ahead AND no wall below:  
    move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — At a fork (path on the right OR path on the left), the agent should **always turn right** if it can, otherwise turn left.

```
if no wall on left:
    turn left
otherwise if no wall on right:
    turn right
```

Hint: the order is wrong.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Solve the maze with forks using **AND** and **OR** conditions. When you finish, come back here.

Send a photo or video of the agent reaching the end, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

I used **AND** when ...

I used **OR** when ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while the agent runs the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **AND**, **OR**, **condition**, **both**, **either**.

1. Show the start of the maze, then run your code.
2. Read each `if ... AND ...` and `if ... OR ...` block out loud and say what makes it run.
3. Show one bug you hit and how you fixed it.
4. Say in your own words how **AND** is different from **OR**.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.